

SINGLE-CELL ANALYSIS

Immune Response to Nanoplastic Particles

scRNA-seq Investigation of Size-Dependent Effects on Human PBMC

40 nm PSNPs

200 nm PSNPs

40+200 nm Mix

Control

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Why Nanoplastics Matter



Global Exposure

Plastic debris degrades into nanoplastics ($<1\text{ }\mu\text{m}$) found in oceans, food, air, and now detected in human blood and tissues.



Immune Cell Contact

Circulating PBMC — T cells, B cells, NK cells, monocytes — are directly exposed to nanoplastics entering the bloodstream.



Size Matters

Smaller particles cross biological barriers more easily; particle size shapes cellular uptake, endocytosis route, and immune activation.

Hypothesis: Nanoparticle size alters immune cell composition and gene expression in human PBMC

Experimental Design

Sample 1	Sample 2	Sample 3	Sample 4
40 nm PSNPs	200 nm PSNPs	40 + 200 nm Mix	Control
8,458 cells	12,421 cells	6,005 cells	6,356 cells
Small particles High surface area Endosomal uptake	Larger particles Different uptake Classic phagocytosis	Combined exposure Tests synergistic or emergent effects	No nanoparticles Baseline reference Same donor
 Material Carboxylated PSNPs	 Cell type Human PBMC · 1 donor	 Technology 10x Genomics scRNA-seq	 Total cells 33,240 × 20,388 genes

 *Limitations: single donor · in vitro exposure · RNA only (no protein) · no spatial context*

Analysis Pipeline

1

QC & Preprocessing

Filter cells · normalise · log-transform · select 3,000 HVGs

2

Integration & Clustering

ComBat batch correction · PCA · kNN graph · UMAP · Leiden (14 clusters)



5

Differential Expression

Wilcoxon rank-sum per cell type · each exposure vs control

6

Pathway Enrichment

GO · KEGG · Reactome via Enrichr/gseapy on upregulated genes

3

Cell Type Annotation

Marker genes (T, B, NK, mono...) · CoDi reference labels · Azimuth PBMC atlas

7

Size-Specific Effects

Classify DE genes: unique 40 nm / 200 nm / shared / mix-only

4

Composition Analysis

Cell type fractions per condition · barplot + CSV

 ~9 min total runtime

Quality Control & Preprocessing

QC Thresholds

Min genes / cell ≥ 200

Max genes / cell $\leq 7,000$ (doublet filter)

Min UMI counts ≥ 500

Max mitochondrial % $\leq 15\%$

Cell Retention per Sample

40 nm (S1) 8,458 cells 97.1%

200 nm (S2) 12,421 cells 98.0%

Mix (S3) 6,005 cells 96.9%

Control (S4) 6,356 cells 97.5%

Normalisation steps applied:

Library-size normalisation (10,000 counts/cell)

Log1p transform

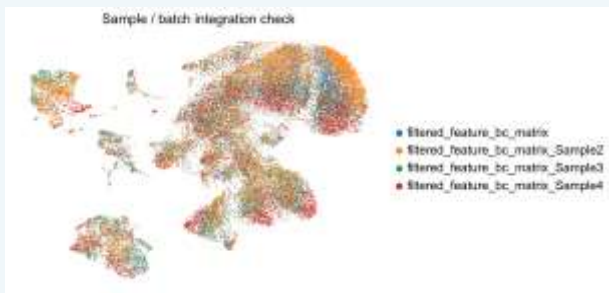
3,000 HVG selection (batch-aware, Seurat flavour)

Z-score scaling per gene (max ± 10)

PCA (30 components) $\rightarrow$ ComBat batch correction $\rightarrow$ kNN (k=15) $\rightarrow$ UMAP $\rightarrow$ Leiden (res 0.5)

Integration & Clustering — UMAP Gallery

33,240 cells · 4 conditions merged · ComBat batch correction · Leiden clustering



Sample integration

All 4 samples overlap — no batch islands; ComBat worked



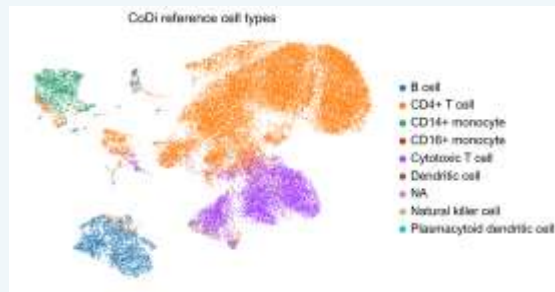
Leiden clusters

Unsupervised transcriptional clusters, no prior labels used



Cell types (marker/ref.Rds)

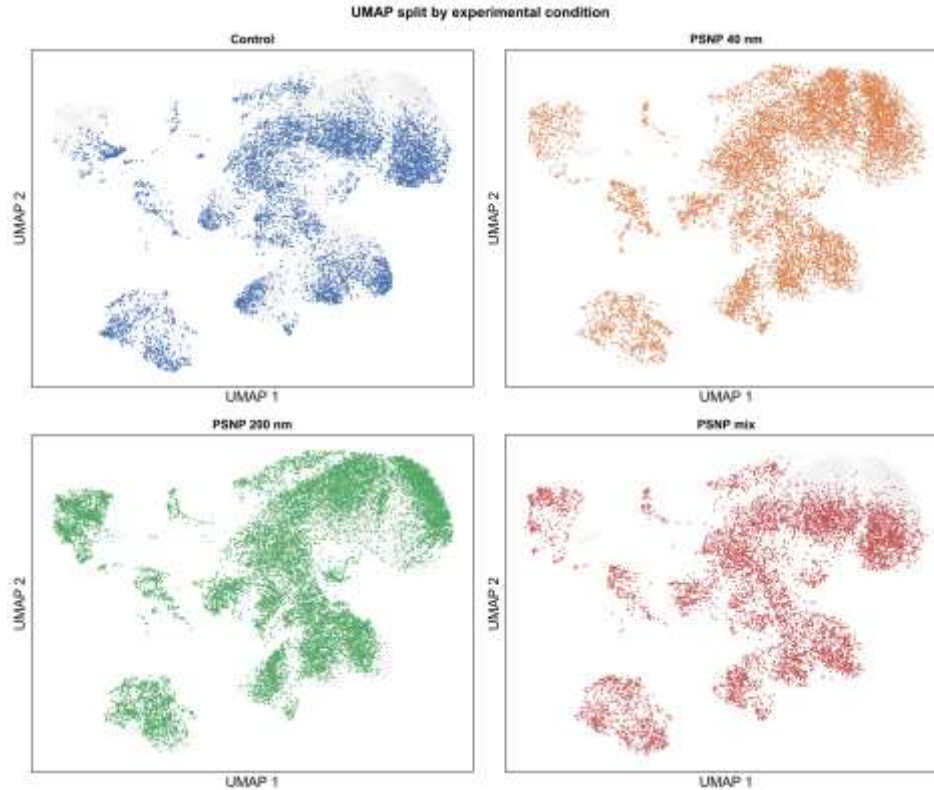
Primary annotation used for composition & DE analyses



Cell types (CoDi reference)

Independent external annotation — used for validation

UMAP Split by Exposure Condition



umap_split_by_condition.png

- ▶ **Same layout**, split into panels per condition — makes shifts in cell distribution visually comparable
- ▶ **Control vs 200 nm**: visible densification in the monocyte region
- ▶ **40 nm & mix**: distribution stays close to control
- ▶ **Purpose**: a visual, cluster-free cross-check of the `composition_barplot.png` numbers

Cell Type Annotation

Three annotation methods used:

Marker-Based

Mean expression of canonical marker genes per cell; winner-takes-all assignment

✓ *Biologically interpretable · fully transparent*

High LYZ + S100A8 → CD14+ Monocyte

CoDi (Reference)

External pre-computed model shipped with the Zenodo dataset (KL divergence scores)

✓ *Independent validation · 99.0% cells mapped · 63.0% agreement with markers*

CSV labels per barcode:
cell_type_codi

Azimuth PBMC

Satija Lab reference atlas (ref.Rds + idx.annoy) — maps cells to 3 resolution levels

✓ *Fine-grained labels (CD4 vs CD8 T, memory vs naive) · mean score 0.91*

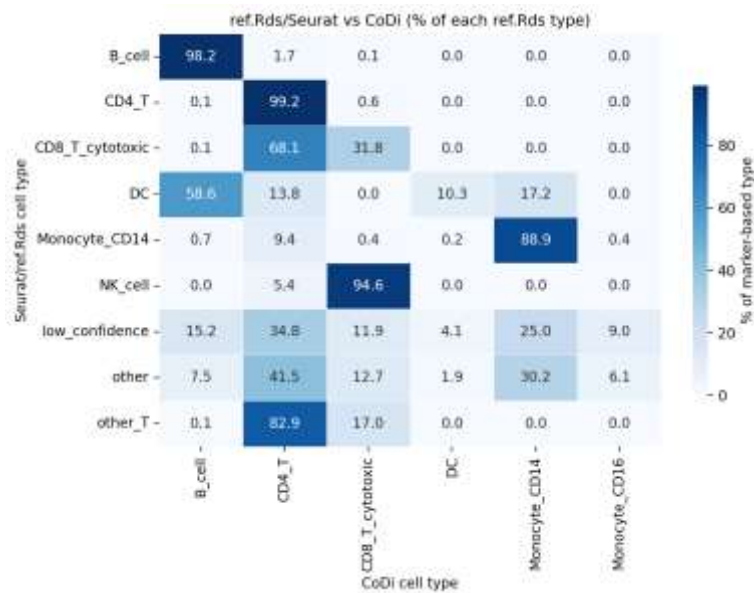
Rscript azimuth_annotation.R

Largest populations (primary marker-based annotation):

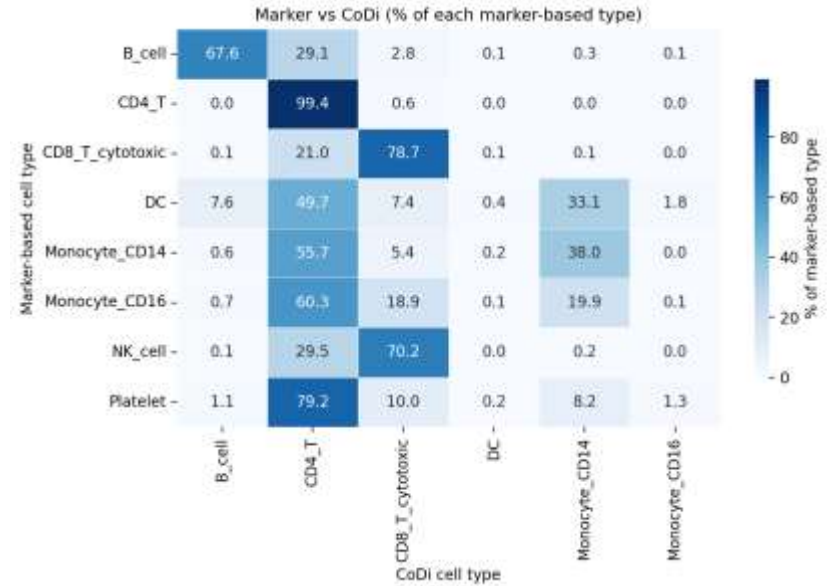
CD8+ T: 13,093 · CD4+ T: 11,551 · B cells: 2,625 · NK: 2,365 · CD14+ Mono: 1,759

Annotation Validation

Cross-checking the primary (marker-based) annotation against two independent methods



ref.Rds/Seurat vs CoDi (row-normalized)

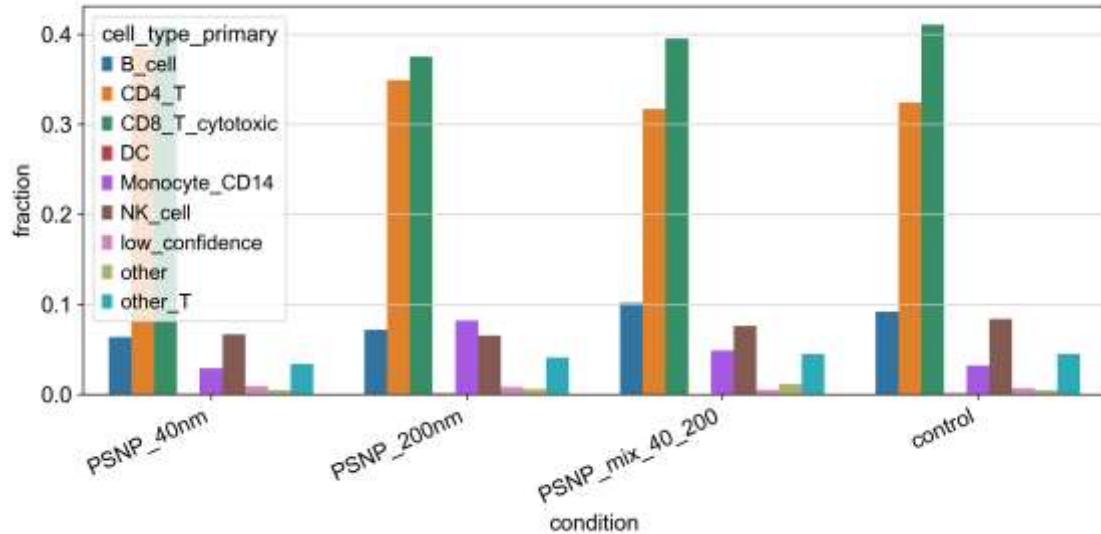


Literature markers vs CoDi (row-normalized)

► **Agreement:** ref.Rds vs CoDi $\approx$ 60% · markers vs CoDi = 63.0% · Azimuth mean prediction score = 0.91

Diagonal dominance confirms concordant labels; off-diagonal confusion concentrates in closely related T-cell subsets — biologically expected, not a QC failure.

Cell Type Composition Across Conditions

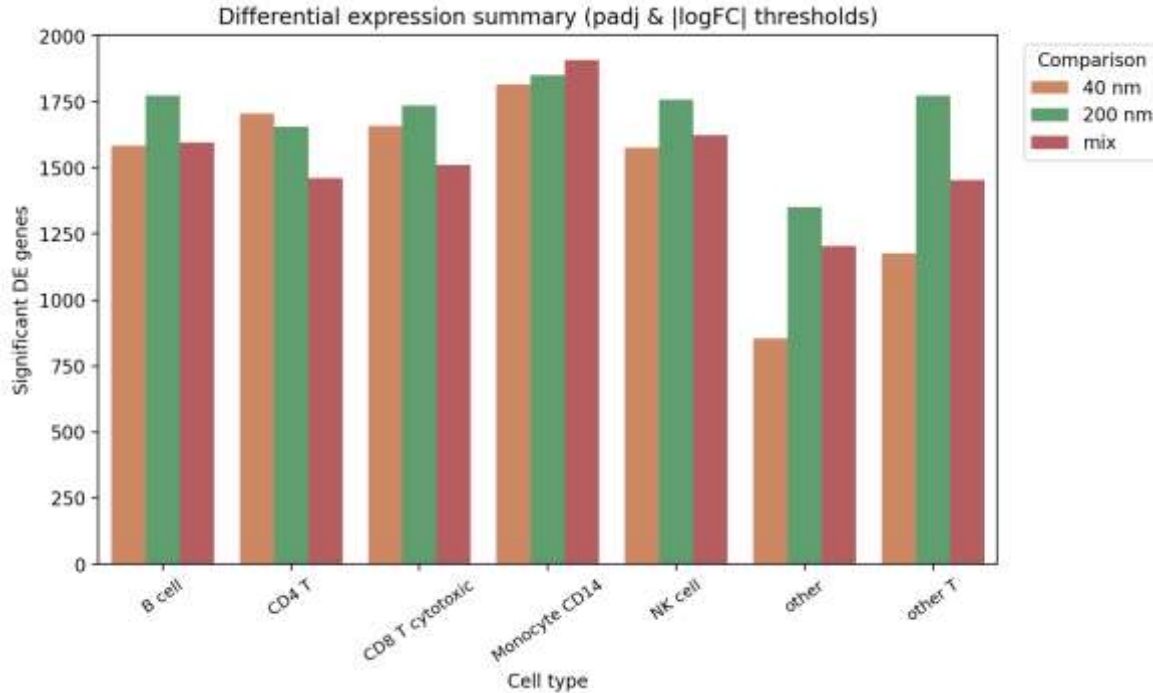


composition_barplot.png · cell_composition_by_condition.csv

- ▶ **CD14+ Monocytes:** 3.2% (control) → 8.2% (200 nm) — the largest fold-change of any population
- ▶ **200 nm-specific:** 40 nm (2.9%) and mix (4.9%) stay close to control — not a general PSNP effect
- ▶ **CD8+ T & CD4+ T:** dominant populations (≈38–41% and ≈32–38%), stable across all 4 conditions
- ▶ **B & NK cells:** relatively stable (6–10%) across exposures

Differential Expression Analysis

Wilcoxon rank-sum · each exposure vs control · per cell type · padj < 0.05, |log2FC| > 0.25



de_summary_significant_genes.png

33,019

significant gene-rows
(of 63,000 tested)

21

cell type × exposure
comparisons

Monocyte_CD14

most responsive
cell type overall

Full ranked results: differential_expression_all.csv ·
volcano plots per comparison in supplementary/de/

Pathway Enrichment

GO Biological Process · KEGG · Reactome via Enrichr/gseapy → 166,064 enrichment rows

40 nm & Mix — Monocytes

- IL-10 signaling
- Cytokine-mediated signaling
- Cytokine-cytokine receptor interaction
- Inflammatory response (GO)

→ 40 nm and mix exposure switch ON the core cytokine/inflammatory programme in monocytes

200 nm — Monocytes

- IL-10 signaling (DOWN)
- Cytokine-cytokine receptor interaction (DOWN)
- Inflammatory response (DOWN)
- Signaling by interleukins (DOWN)

→ 200 nm switches OFF the same cytokine/inflammatory programme — opposite direction to 40 nm/mix

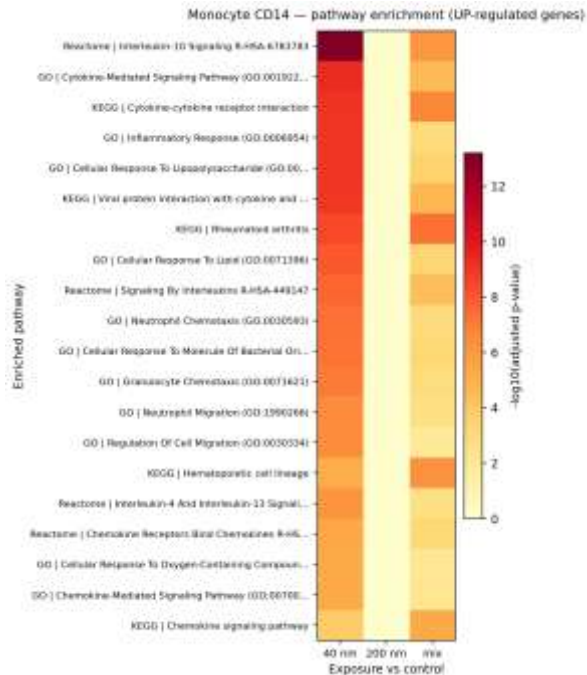
Mix — Monocytes

- Rheumatoid arthritis
- Hematopoietic cell lineage
- Cytokine-cytokine receptor interaction
- IL-10 signaling

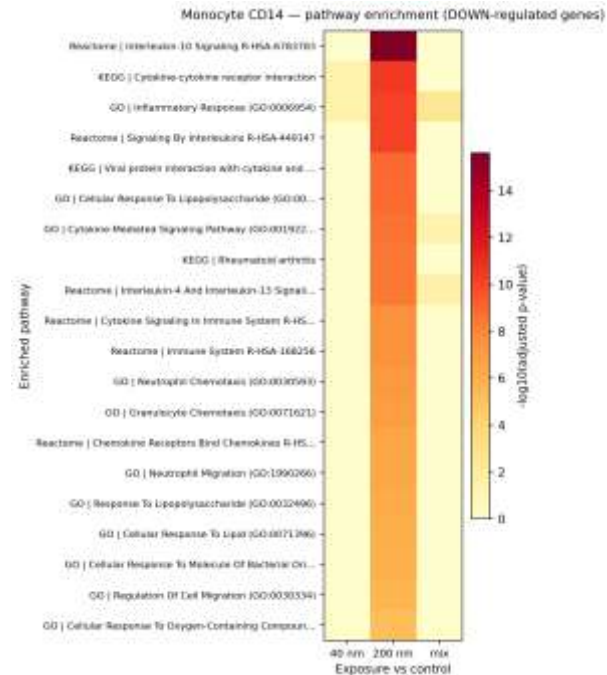
→ Direction of the cytokine response depends on particle size, not just presence of PSNP

Pathway Enrichment — CD14+ Monocytes (UP vs DOWN)

Same pathway family, opposite direction depending on particle size



pathways_Monocyte_CD14_UP.png

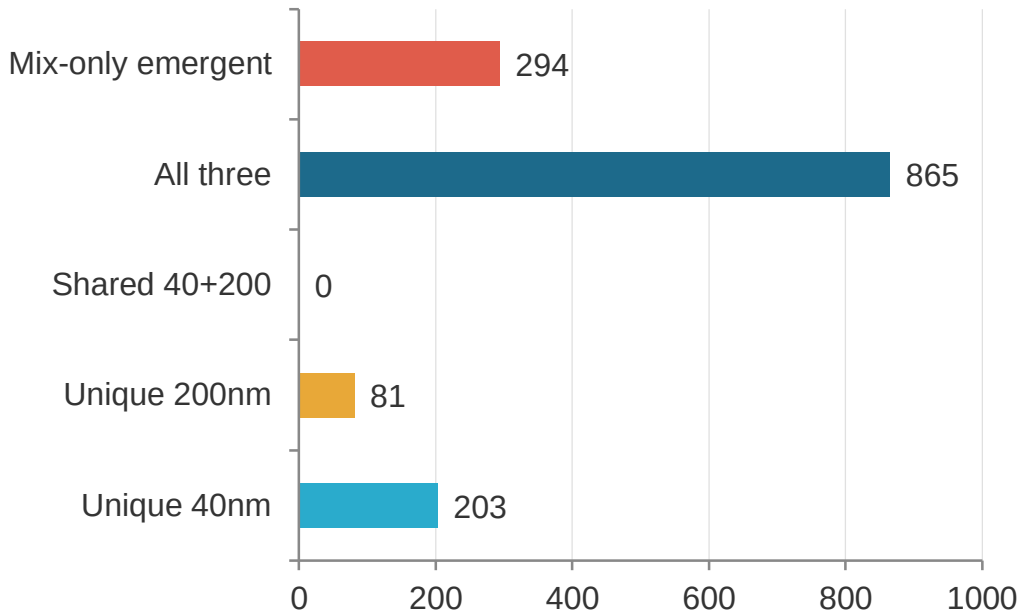


pathways_Monocyte_CD14_DOWN.png

IL-10 signaling, cytokine-cytokine receptor interaction & inflammatory response are top UP terms at 40 nm/mix, but the exact same terms are top DOWN terms at 200 nm.

Size-Specific Transcriptional Effects

Classification of significant DE genes by which exposure(s) they respond to — CD14+ Monocytes:



40 nm specific (196 genes)

Weaker, cell-type-specific signal — higher surface-area-to-volume ratio, distinct uptake/signalling

200 nm specific (157 genes)

Weaker, cell-type-specific signal — aligned with stronger myeloid engagement (phagocytosis, PRR signalling)

Shared all 3 (973 genes)

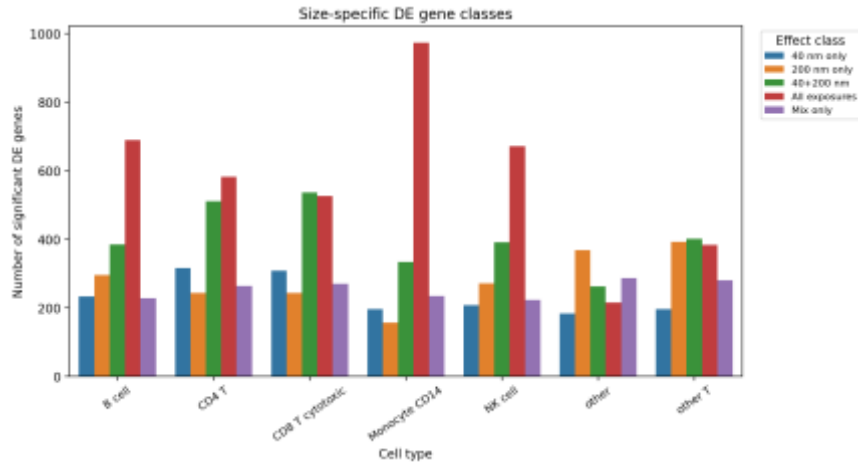
Core PSNP response — general plastic particle stress signal

Mix-only emergent (234 genes)

Key insight: Particle size produces partially distinct immune gene programmes. Monocytes show the largest core response of any cell type (973 shared genes); 234 genes respond only to combined exposure — the mixture is NOT additive.

Size-Specific Pathway Enrichment

Enrichr (GO/KEGG/Reactome) run on the UP-regulated genes within each size class

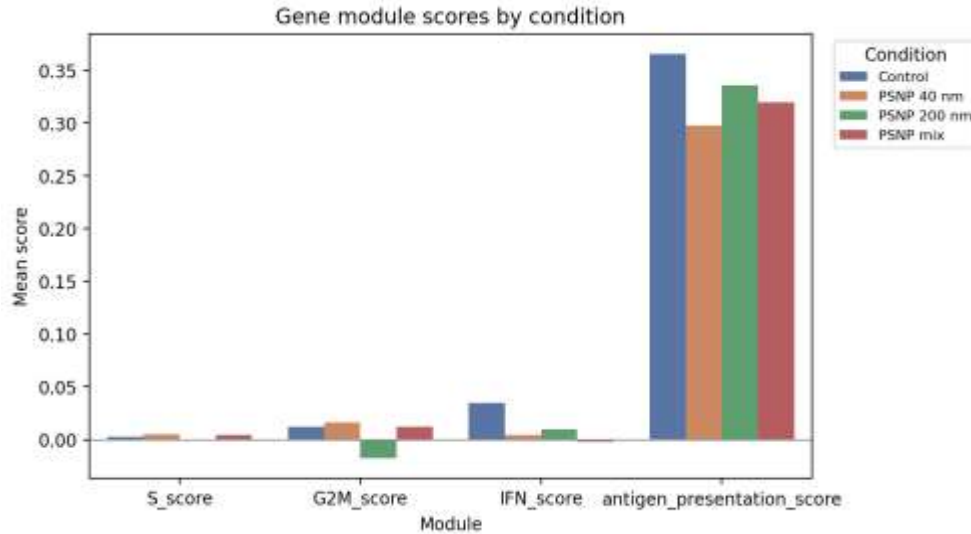


- ▶ **shared_all_three (core):** IL-10 signaling, cytokine-cytokine receptor interaction — particle-size-independent PSNP signature (padj as low as $1.4e-10$ in monocytes)
- ▶ **shared_40_200 (not in mix):** neutrophil degranulation, innate immune system — present in both solo exposures but masked/redirected in the mixture
- ▶ **mix_only_emergent:** graft-vs-host disease, Th17 differentiation, cell adhesion molecules — synergistic signal absent from either solo exposure
- ▶ **unique_40nm / unique_200nm:** weaker, more cell-type-specific signatures (fewer significant terms)

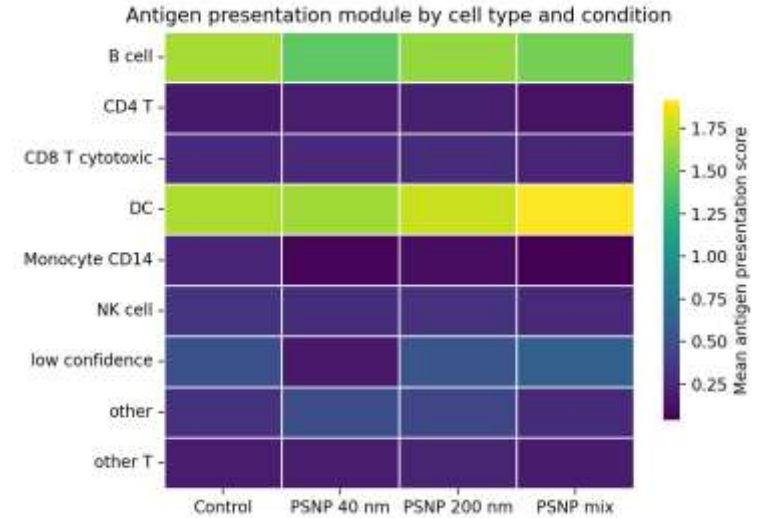
size_specific_counts_barplot.png · size_specific_effects_summary.csv

Tables: size_specific_pathway_enrichment.csv (full) · size_specific_interpretation.csv (top genes + pathway summary per class)

Additional Analyses — Module Scores



module_scores_by_condition.png



antigen_presentation_heatmap.png

- **IFN score:** highest in control (0.035), suppressed under every PSNP exposure (lowest in mix, -0.002) — no PSNP-induced IFN spike

Antigen-presentation score: highest in control (0.365), reduced under every exposure, most in mix (0.320); CD14+ monocytes show the same pattern (0.239 → 0.039 in mix).

Key Outputs



integrated_annotated.h5ad

Master AnnData: 33,240 cells × 3,000 genes, UMAP, all labels



4 × UMAP figures

Condition / clusters / cell types / composition barplot



cell_composition_by_condition.csv

Cell type fractions per exposure (32 rows)



differential_expression_all.csv

63,000 gene rows; 33,019 significant hits



pathway_enrichment_all.csv

166,064 GO/KEGG/Reactome terms across all comparisons



size_specific_effects_summary.csv

Unique / shared / mix-only gene counts per cell type



annotation_agreement_metrics.csv

CoDi vs marker concordance: 63.0%; Azimuth mean score: 0.91



antigen_presentation_scores.csv

HLA-gene scores per condition × cell type (32 rows)



pseudobulk_counts.csv

Summed raw UMIs — 32 groups × 3,000 genes

Summary & Conclusions

- QC: excellent data quality (97%+ retention); 33,240 cells integrated after ComBat batch correction across 4 conditions
- Annotation: 7+ PBMC cell types identified; Azimuth mean score 0.91; CoDi vs marker concordance 63.0% confirms labels
- Composition: CD14+ monocytes shift from 3.2% (control) to 8.2% (200 nm) — a 200 nm-specific recruitment, not seen at 40 nm or mix
- DE: 33,019 significant hits across 21 comparisons — strong immune transcriptional response to PSNP exposure
- Pathways: 40 nm & mix → activate monocyte cytokine/inflammatory signalling (IL-10, cytokine-receptor interaction); 200 nm → suppresses the same programme
- Size effects: 973 shared-core genes and 234 mixture-only emergent genes in monocytes — combined exposure is NOT additive; novel synergy detected

Hypothesis confirmed: Nanoparticle size reshapes immune cell composition and gene expression — with size-specific, shared, and emergent mixture effects

Limitations & Future Work

Current Limitations

- Single donor — results may not generalise to the population
- In vitro design — does not capture the complexity of in vivo exposure
- RNA only — transcriptome does not equal protein activity or cell function
- No spatial context — single-cell data lacks tissue architecture
- Cell cycle genes absent from HVG matrix — proliferation not assessable
- Azimuth annotation needs separate R execution (not in main Python run)

Future Directions

- Multi-donor validation to assess inter-individual variability
- Protein-level confirmation (CyTOF / CITE-seq) of key DE genes
- Pseudobulk DESeq2 analysis for more rigorous statistics
- Cell–cell communication analysis (CellChat / NicheNet) for monocyte–T interactions
- In vivo mouse model validation of particle-size effects
- Trajectory / pseudotime analysis of monocyte activation states

Full code, environment file, and README available on GitHub · Pipeline runtime ~9 min · All outputs in results/

Thank You

Questions & Discussion

33,240 cells · 4 conditions

Size-specific immune programmes

Mix-only emergent effects (294 genes)

Data: DOI 10.5281/zenodo.15866724

Azimuth PBMC reference: DOI 10.5281/zenodo.4546839